Deployment Paper

California Housing Price Prediction Using Machine Learning

Student Information

- **Project Title:** California Housing Price Prediction
- **Project Type:** Machine Learning (Regression)
- **Language:** Python
- **Dataset:** California Housing Dataset

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1. Introduction

The real estate industry relies on accurate house price estimation for buying, selling, and investment decisions. This project uses the California Housing dataset to analyze housing characteristics and estimate the median house value using machine learning techniques. The project focuses on data exploration, understanding relationships between variables, and building a predictive model.

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2. Problem Statement

Estimating house prices manually can be difficult because many factors influence property values, such as income, location, population, and housing characteristics.

The objective of this project is to build a machine learning model that predicts the median house value based on available housing features.

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3. Research Question

Can machine learning accurately estimate California house prices using housing-related features?

This question aims to support buyers, sellers, and real estate analysts in making informed pricing decisions.

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4. Objectives

- Analyze the California Housing dataset.
- Understand important housing features.
- Perform exploratory data analysis.
- Build a regression model for house price prediction.
- Evaluate model performance.
- Provide decision-support based on observed patterns.

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5. Dataset Description

The California Housing dataset contains information collected from California districts.

Features

Feature	Description
longitude	Geographic longitude
latitude	Geographic latitude
housing_median_age	Median age of houses
total_rooms	Total number of rooms
total_bedrooms	Total bedrooms
population	Population of the district
households	Number of households
median_income	Median income
median_house_value	Target variable

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6. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook

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7. Methodology

The project followed these steps:

1. Load the California Housing dataset.
2. Explore the dataset.
3. Check missing values.
4. Generate summary statistics.
5. Analyze housing features.
6. Prepare data for machine learning.

7. Train a regression model.
8. Evaluate prediction performance.

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8. Exploratory Data Analysis

The analysis included:

- Number of records
- Number of features
- Missing value detection
- Average house price
- Maximum house price
- Average median income

Example observations:

- Dataset contains multiple housing records.
- No major missing values were found.
- Median income appears to influence house prices.
- House values vary significantly across districts.

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9. Decision and Business Impact

Decision Supported

Estimate the market value of residential houses.

Users

- Home buyers
- Home sellers
- Real estate companies
- Property investors

Cost of Wrong Prediction

- Buyers may overpay.
- Sellers may lose money.
- Investors may make poor investment decisions.

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10. Results

The exploratory analysis successfully identified important housing characteristics.

The project provides:

- Housing data insights
- Statistical summaries
- Decision-support for price estimation

The model does **not** guarantee future prices but provides useful estimates based on historical data.

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11. Limitations

This project:

- Uses historical data only.
- Cannot predict future economic conditions.
- Does not establish cause-and-effect relationships.
- Should not be considered financial advice.

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12. Future Improvements

Future work may include:

- Testing multiple regression algorithms.
- Hyperparameter tuning.
- Feature engineering.
- Cross-validation.
- Deploying the model as a web application using Streamlit or Flask.

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13. Conclusion

The California Housing Price Prediction project demonstrates how machine learning can support real estate decision-making. By analyzing housing features and historical data, the model estimates median house values and provides valuable insights. While predictions are not guaranteed, they can help buyers, sellers, and analysts make better-informed decisions.

References

1. Scikit-learn Documentation
2. Pandas Documentation
3. NumPy Documentation

4. California Housing Dataset